Discrete Structures

Topic 3: Combinatorics (Counting)

3 · Why Do We Count?

- **Combinatorics is the art of counting without listing everything.**

  Listing 1,000,000 passwords one by one is impossible. A formula gives the answer in seconds.

- **Computer science uses counting to find:**

  How many passwords are possible?

  How many routes are there in a network?

  How many steps will an algorithm take?

- **Two big questions: Does order matter? Can items repeat?**

3 · Product Rule (AND)

- **If a job has 2 steps, and step 1 can be done in m ways and step 2 in n ways, then the job can be done in m × n ways.**

  Why? For EACH of the m first choices, there are n second choices. So we get n + n + ... (m times) = m × n.

- **Example: 3 shirts and 2 pants.**

  Each shirt goes with 2 pants, so 3 × 2 = 6 outfits.

- **Remember: AND means MULTIPLY.**

3 · Example: Product Rule

A 4-digit PIN, digits 0–9, repeats allowed:

$10 \times 10 \times 10 \times 10 = 10{,}000$

Each of the 4 places has 10 choices.

A PIN with NO repeated digit:

$10 \times 9 \times 8 \times 7 = 5{,}040$

After using one digit, one fewer digit is left for the next place.

Plate: 2 letters then 3 digits: $26 \times 26 \times 10 \times 10 \times 10 = 676{,}000$

3 · Sum Rule (OR)

- **If a job can be done in m ways OR in n ways, and both cannot happen together, then the total is m + n.**

  Why? We choose only ONE thing, so we just count all the options together.

- **Example: Choose one dessert: 4 cakes or 3 ice creams.**

  4 + 3 = 7 choices.

- **Remember: OR means ADD.**

  Product rule: do step 1 AND step 2. Sum rule: do this OR that.

3 · Example: Both Rules Together

Go from city A to city C.

Route 1: via B. There are 3 roads A to B and 2 roads B to C.

Both parts are needed (AND): 3 × 2 = 6 routes.

Route 2: a direct road. There are 4 direct roads.

4 routes.

Either Route 1 OR Route 2:

6 + 4 = 10 routes in total.

3 · Factorial and Arranging Objects

- **$n! = n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1$**

  $4! = 4 \times 3 \times 2 \times 1 = 24$

- **Number of ways to arrange n different objects = n!**

  Why? n choices for the first place, then n−1 for the second, and so on (product rule).

- **Example: arrange A, B, C: 3! = 6**

  ABC, ACB, BAC, BCA, CAB, CBA

- **0! = 1**

  There is exactly one way to arrange nothing.

3 · Permutations: Order Matters

- **Choose r objects from n and put them in order.**
- **P(n, r) = n! / (n − r)! = n × (n−1) × ... (r numbers)**

  Why? n choices for place 1, n−1 for place 2, ... The division by (n−r)! removes the extra tail of the factorial.

- **Example: 10 runners. Gold, silver, bronze medals.**
- **P(10, 3) = 10 × 9 × 8 = 720**

  Order matters: Ali gold and Sara silver is different from Sara gold and Ali silver.

3 · Example: Letters With Repeats

How many different words can we make from the letters of BOOK?

If all 4 letters were different: 4! = 24.

But the two O's are the same.

Swapping the two O's does not give a new word. Each word was counted 2 times.

So divide by 2!: 4! / 2! = 12

General rule: n! / (a! × b! × ...) where a, b are the counts of repeated letters.

3 · Combinations: Order Does Not Matter

- **Choose r objects from n as a group.**
- **$C(n, r) = n! / (r! \times (n - r)!)$**

  Why? P(n, r) counts every group r! times (once for each order inside the group). So divide by r!.

- **Example: choose 3 students from 5.**

  P(5,3) = 60. Divide by 3! = 6. So C(5,3) = 10.

- **$C(n, r) = C(n, n-r)$**

  Choosing who is IN is the same as choosing who is OUT.

3 · Permutation or Combination?

- **Ask: does the ORDER matter?**
- **YES: use Permutation P(n, r)**

  President and vice-president, race positions, passwords, arranging books.

- **NO: use Combination C(n, r)**

  Committee, team, hand of cards, choosing a set of friends.

- **Quick test: if swapping two chosen items gives a NEW result, order matters.**

3 · Example: Combinations With AND and NOT

From 5 men and 4 women, choose 2 men and 2 women.

C(5,2) × C(4,2) = 10 × 6 = 60

Both choices are needed (AND), so multiply.

Choose a team of 3 with AT LEAST one woman.

Total teams: C(9,3) = 84. Teams with no woman: C(5,3) = 10.

Answer: 84 − 10 = 74

"At least one" is easier as: ALL minus NONE.

3 · Pascal's Triangle

Row 0: 1

Row 1: 1 1

Row 2: 1 2 1

Row 3: 1 3 3 1

Row 4: 1 4 6 4 1

Each number = the two numbers above it added. Row n lists C(n, 0), C(n, 1), ..., C(n, n).

Pascal's rule: $C(n, r) = C(n-1, r-1) + C(n-1, r)$

3 · Binomial Theorem

- **$(x + y)^n = C(n,0)x^n + C(n,1)x^{n-1}y + \ldots + C(n,n)y^n$**

  Why? Each term picks x or y from each of the n brackets. C(n,k) counts the ways to pick y exactly k times.

- **Example: $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$**

  The coefficients 1, 3, 3, 1 are Row 3 of Pascal's triangle.

- **Coefficient of x^2y^3 in $(x + y)^5$ is $C(5,3) = 10$.**
- **Sum of all coefficients = 2^n (put $x = 1$, $y = 1$).**

3 · Summary

- **Product rule: AND → multiply. Sum rule: OR → add.**
- **n! = ways to arrange n different objects.**
- **P(n, r) = n!/(n−r)! when ORDER matters.**
- **C(n, r) = n!/(r!(n−r)!) when order does NOT matter.**
- **Repeated letters: divide by the factorials of the repeat counts.**
- **"At least one" = total − none.**

3 · Practice Questions

Q1. A cafe has 4 sandwiches, 3 drinks and 2 desserts. A meal has one of each. How many meals?

Q2. How many 3-letter codes can be made from A–Z with no repeated letter?

Q3. How many different arrangements of the letters of LEVEL?

Q4. Choose a team of 4 from 9 players. How many ways?

Q5. 6 boys and 5 girls. Choose a committee of 4 with exactly 2 girls.

Q6. Find the coefficient of x^2y^2 in $(x + y)^4$.

3 · Solutions Q1 and Q2

Q1: Three choices, all needed (AND).

4 × 3 × 2 = 24 meals

Multiply because one of each is chosen.

Q2: Order matters (ABC is a different code from CBA) and no repeats.

26 × 25 × 24 = 15,600

26 choices, then 25, then 24, since used letters are gone. Same as P(26, 3).

3 · Solutions Q3 and Q4

Q3: LEVEL has 5 letters: L twice, E twice, V once.

5! / (2! × 2!) = 120 / 4 = 30

Divide by 2! for the L's and 2! for the E's, since swapping equal letters gives the same word.

Q4: A team is a group, so order does not matter.

C(9, 4) = 9! / (4! × 5!) = (9×8×7×6) / 24 = 126

3024 / 24 = 126

3 · Solutions Q5 and Q6

Q5: Exactly 2 girls and so 2 boys.

Girls: C(5, 2) = 10. Boys: C(6, 2) = 15.

Both choices are needed (AND), so multiply: 10 × 15 = 150

Q6: The term with x^2y^2 has coefficient C(4, 2).

C(4, 2) = 4! / (2! × 2!) = 6

Check with Row 4 of Pascal's triangle: 1, 4, 6, 4, 1. The middle number is 6 ✓